criterion: {'gini', 'entropy'}, default = 'gini'

Gini Impurity

$$E = -P_y \log_2(P_y) - P_n \log_2(P_n)$$

$$G_I = 1 - (P_y^2 + P_N^2)$$

- Impurity of child col^ns:

~~Higher~~ Pure = low entropy = low gini impurity

IG → entropy
→ gini

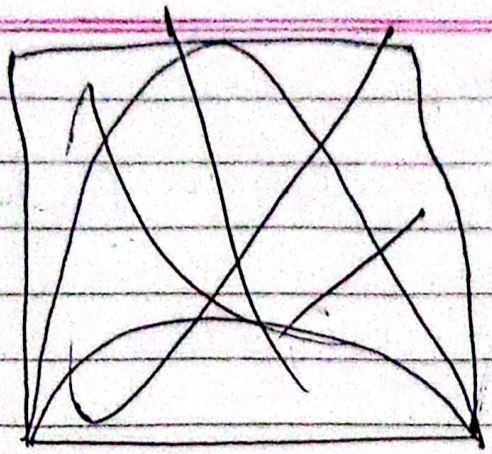
~~IG~~ = Diff:

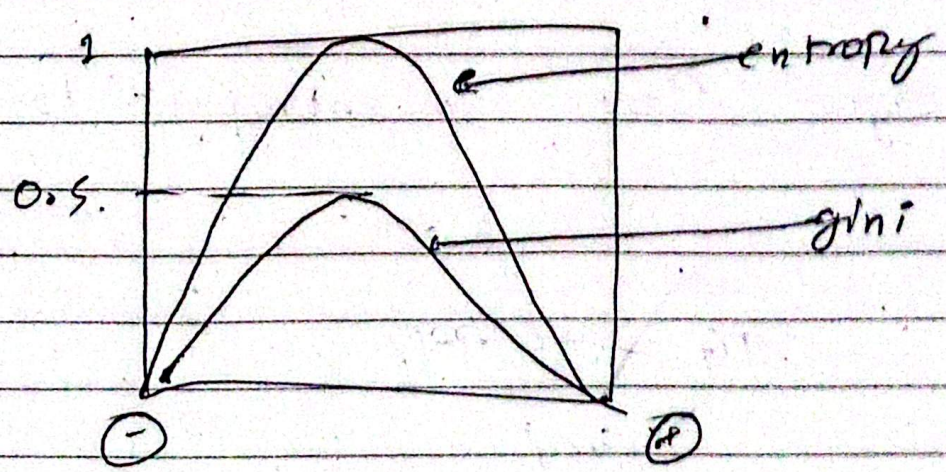
$P_y = 1$
$P_y = 0$ → $E = 0$.

→ $G.I = 0$

$P_y = 0.5$
$P_N = 0.5$ → $E = 1$

→ $G.I = 0.5$





- Gini is faster
  (cal sq. faster than cal. log.)

Sometimes,

- Entropy gives balance tree, but gini impurity gives overfitting.

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Decision Tree in numerical.

User Rating	3.5	4.1	5	4.4
Download	Yes	NO	Yes	Yes

~~Rating~~

Step1: Sort data in ascending.
Step2: Split the entire data on the basis of ~~numerical~~ every value of user_rating.

Do (rating > 3.5)

Dataset 1

~~Yes~~ NO
3.5
Yes

Dataset 2

Yes		
4.1	4.4	5
NO	Yes	Yes

Again : (rating > 4.1)

D1

NO	
3.5	4.1
Yes	NO

D2

Yes	
4.4	5
Yes	Yes.

Again : (rating > 4.4)

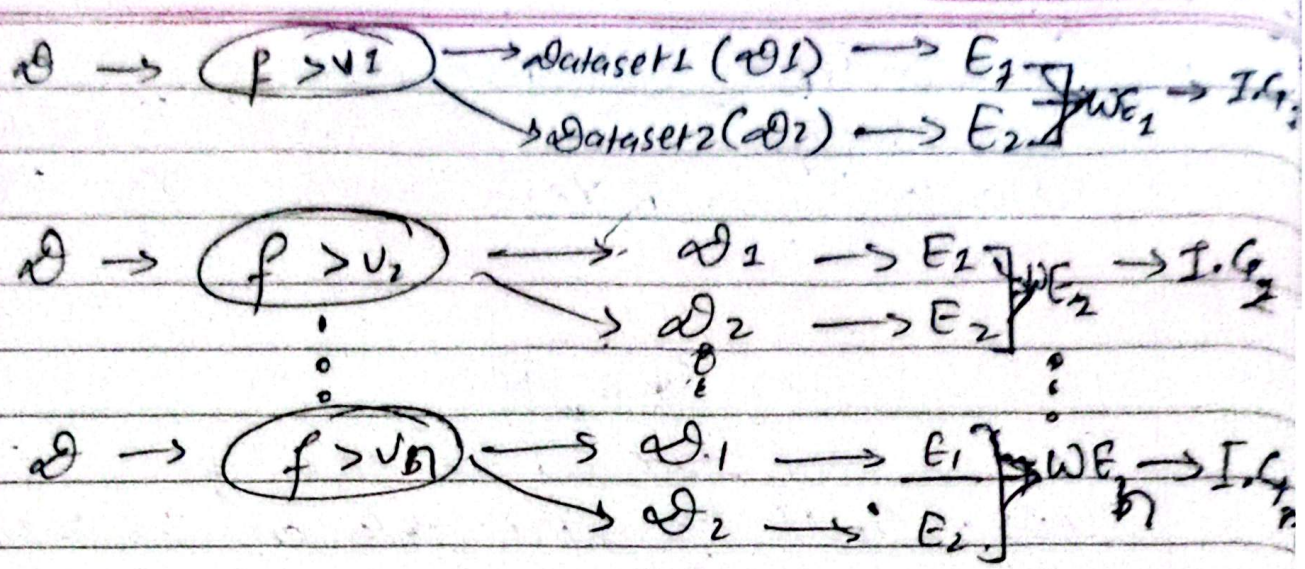
D1

No		
3.5	4.1	4.4
Yes	NO	Yes

D2

Yes
5
Yes

- Splitting every in two parts.



- Steps:
  Max[IG1, IG2, - - - IGn].

- Do this till leaf nodes.

- It takes a lot of time, but we have to do this only once in training phase.

- If test data comes, it doesn't do computation at that time. So,
  ∴ Test time complexity only log

  So, faster for categorical data.
  try to avoid in numerical.

- So overfitting occurs, better in train but not in test